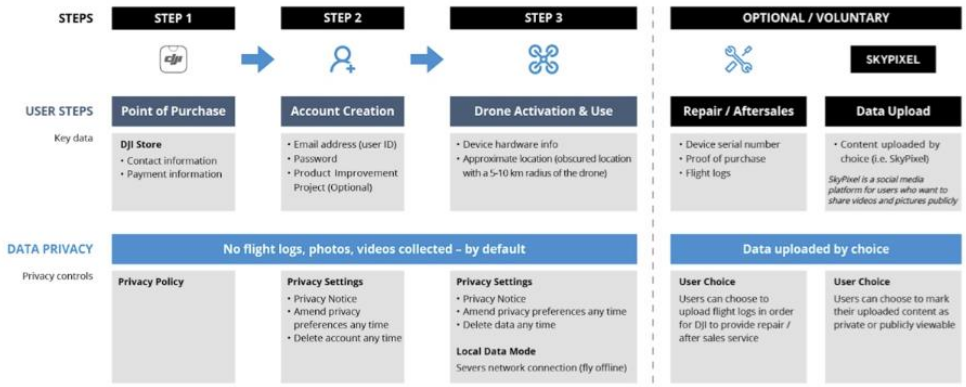
Infringement Chart Strength: High

DJI Fly App release: 2019

US10942026B2	DJI Fly App
18. A method for collaboratively utilizing aerial vehicle data for at least one unmanned aerial vehicle, comprising the steps of:	*DJI Fly allows a user to utilize the aerial vehicle data (such as live feed or video etc.) for at least one unmanned aerial vehicle (e.g., DJI Avata 2, Dji Mini 4 Pro etc.)* DJI FLY **Your Versatile Flight Companion** Built to optimize your aerial experience, the DJI Fly app empowers users to fly, capture incredible images, edit, and share with ease. The app interface was designed to be simplified and ultra-intuitive. Thanks to comprehensive flight tutorials and the easy-to-use editor, you can now use your FlyCam right out of the box and create stunning content with confidence. Source: https://www.dji.com/global/dji-fly **DJI Fly** The DJI Fly app interface was designed to be simplified and ultra-intuitive. Compatible for DJI Avata 2, DJI Mini 4 Pro, DJI Air 3, DJI Mavic 3 Pro, DJI Mini 2 SE, DJI Mini 3, DJI Mavic 3 Classic, DJI Avata, DJI Mini 3 Pro, DJI Mavic 3, DJI Mini SE, DJI Air 2S, DJI FPV, DJI Mini 2, Mavic Air 2, Mavic Mini. Source: https://www.dji.com/global/downloads/djiapp/dji-fly

	User Guide: Consumer Drone Privacy Controls The DJI Fly flight app is what enables you to operate your DJI consumer drone. It provides you with real-time live display, flight operation tools, user information management, image editing, and content sharing options. When you activate your drone for the first time, DJI Fly will take you through a short account registration process, outlining what types of data can be collected (including what is optional), and registering your preferences along the way. Source: https://www.dji.com/global/trust-center/resource/consumer-privacy-controls
providing access through a login process into an informed decision loop managed by an unmanned aerial vehicle data management system capable of fusing dynamic data streams;	*DJI Fly app provides access through a login process (e.g., user login) into an informed decision loop (e.g., able to manage and control the operations of the unmanned aerial vehicle) managed by an unmanned aerial vehicle data management system (e.g., DJI management system or cloud) capable of fusing dynamic data streams (e.g., media such as video or images), where a user enter their login credentials to access the services provided by the DJI cloud provider.* **TYPES OF DATA YOU MAY SHARE WITH DJI** **Required:** An email address, a password and the drone serial number (for product activation) **Recommended:** For safety purposes, the drone will attempt to conduct an approximate location check upon activation in order to load the relevant local map and geofencing zones. If desired, you can opt-out of this safety feature by activating "Local Data Mode". To protect user privacy, the DJI flight app applies a random offset within a 5-10 km radius to prevent identification of the operator's actual location. The data collected is used temporarily and not stored on any DJI server.



Source: https://www.dji.com/global/trust-center/resource/consumer-privacy-controls

SECURITY

DJI application software includes DJI Fly series, DJI Pilot series, DJI Agras series, and other software which plays a central role in interacting with drones, displaying the image taken by the drone, controlling the drone, editing images, providing user registration and login, and interacting with cloud services. Therefore, the app is the main component in the drone system.

DJI FLY

The DJI Fly applications interact directly with consumer drone systems. They carry out drone system communication, interactive control, real-time live display, user information management, image editing, and content sharing.

The flight app communicates clearly to users what data can be collected (including what is optional) from point of purchase to drone activation. No flight logs, images or videos are shared with DJI without user consent (see Figure 6).

3. **Cloud infrastructure:** Provides drones with services such as media data sharing, data storage, and remote management, playing an important role in supporting modern drone management.
    - **DJI official cloud services:** The infrastructure which DJI products and services rely on, including DJI Service, SkyPixel, and DJI FlightHub 2.
    - **User private cloud services:** Third-party cloud services that can be accessed by certain DJI products to enhance data confidentiality for enterprise users.

There are three main types of links in DJI device interconnection:

- **Transmission link:** Used for transmitting control and payload data between the air system and the ground control. DJI currently supports two types of transmission links: OcuSync links and 4G enhanced transmission links.
- **Cloud connection link:** Used for communication between the air system, the ground control, and the cloud infrastructure. DJI products currently support connection to DJI official cloud services as well as user private cloud services.
- **QuickTransfer link:** Used for communication between user mobile devices and the air system, allowing quick access to media data on drones and improving user experience. Currently, QuickTransfer links are used mainly on consumer drones.

Source: https://terra-1-g.djicdn.com/851d20f7b9f64838a34cd02351370894/trust%20center/DJI%20Drone%20Security%20White%20Paper.pdf

Introduction to Waypoint Flight Mode of DJI Fly

If you want to know how to use Waypoint Flight, you can refer to this article for reference.

Applicable Products:DJI Fly

Introduction

In Waypoint flight mode, the aircraft can automatically complete a preset flight trajectory and shooting actions according to the planned route parameters.

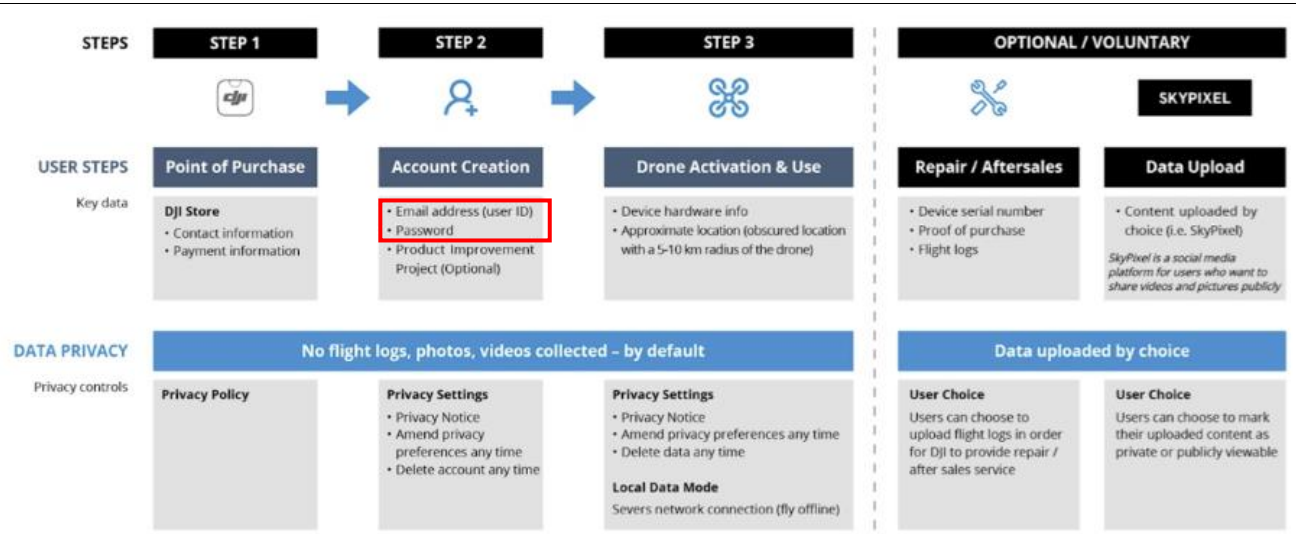
Takeoff Preparations

Run the DJI Fly app and connect to the aircraft. Tao Go Fly to enter the aircraft control view. On the left side of the screen, tap the Waypoint Flight icon to start route planning.



Informed decision loop where a user could create waypoints for the unmanned aircraft

	Make Waypoints Routes can be generated after the Waypoints are marked, and the aircraft will fly along the set route. DJI Mini 4 Pro, DJI Air 3, DJI Mavic 3/ Mavic 3 Cine/Mavic 3 Classic are able to design a flight path by setting points before take-off. 1. After the aircraft takes off, pin waypoints by using the remote controller or operating on the camera view: • Remote control: enter the current Waypoint operation panel, press the remote control Fn button, or the C1 button if you are using DJI RC. • The operation interface: In the Waypoint operation panel, tap the plus sign. • After the Waypoint is marked, the aircraft will record the flight altitude, nose direction, gimbal pitch angle, and other data at this point. Source: https://support.dji.com/help/content?customId=en-us03400007343&spaceId=34&re=US&lang=en&documentType=artical&paperDocType=paper
authenticating at least one user using at least one mobile device pursuant to data management system rights and policies;	*The authenticating (e.g., using credentials provided by the user) at least one user using at least one mobile device (e.g., iPhone or Android, where the DJI app could be installed) pursuant to data management system rights and policies (e.g., DJI cloud management).* **TYPES OF DATA YOU MAY SHARE WITH DJI** **Required:** An email address, a password and the drone serial number (for product activation) **Recommended:** For safety purposes, the drone will attempt to conduct an approximate location check upon activation in order to load the relevant local map and geofencing zones. If desired, you can opt-out of this safety feature by activating "Local Data Mode". To protect user privacy, the DJI flight app applies a random offset within a 5-10 km radius to prevent identification of the operator's actual location. The data collected is used temporarily and not stored on any DJI server.



USER PRIVACY CONTROLS

You can change your privacy preferences at any time by simply toggling the "on" or "off" authorizations of the specified item in the "Settings" page found in the app.

Any data transmitted between the DJI Fly app and the cloud server are encapsulated with the HTTPS protocol to ensure secure transmission. You can also choose to enable the Local Data Mode (LDM) function - which severs the communication between your flight app and the server.

Source: https://www.dji.com/global/trust-center/resource/consumer-privacy-controls

SECURITY

DJI application software includes DJI Fly series, DJI Pilot series, DJI Agras series, and other software which plays a central role in interacting with drones, displaying the image taken by the drone, controlling the drone, editing images, providing user registration and login, and interacting with cloud services. Therefore, the app is the main component in the drone system.

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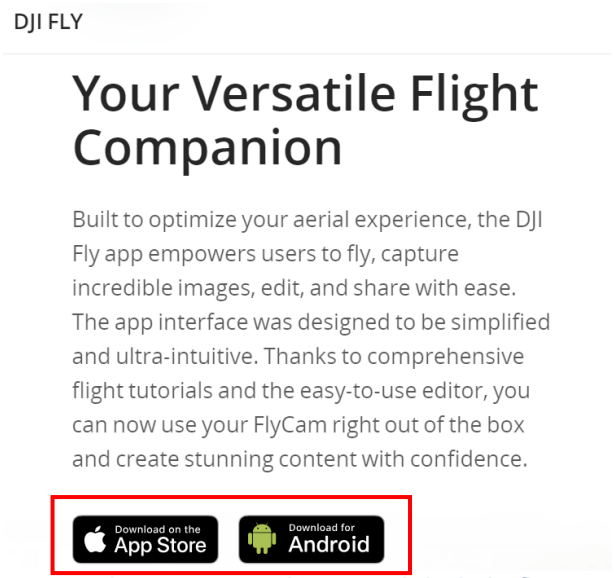
The flight app communicates clearly to users what data can be collected (including what is optional) from point of purchase to drone activation. No flight logs, images or videos are shared with DJI without user consent (see Figure 6).

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Source: https://terra-1-g.djicdn.com/851d20f7b9f64838a34cd02351370894/trust%20center/DJI%20Drone%20Security%20White%20Paper.pdf



Source: https://www.dji.com/global/dji-fly

after authenticating the at least one user, allowing the at least one user to display on the at least one mobile device a map;	*In response to the authentication, the DJI Fly app allows the user to display on the at least one mobile device (e.g., Android or iPhone) a map.* *User utilize the DJI Fly app to access map in the application.*  Source: https://www.dji.com/global/dji-fly

Make Waypoints

Routes can be generated after the Waypoints are marked, and the aircraft will fly along the set route.

DJI Mini 4 Pro, DJI Air 3, DJI Mavic 3/ Mavic 3 Cine/Mavic 3 Classic are able to design a flight path by setting points before take-off.

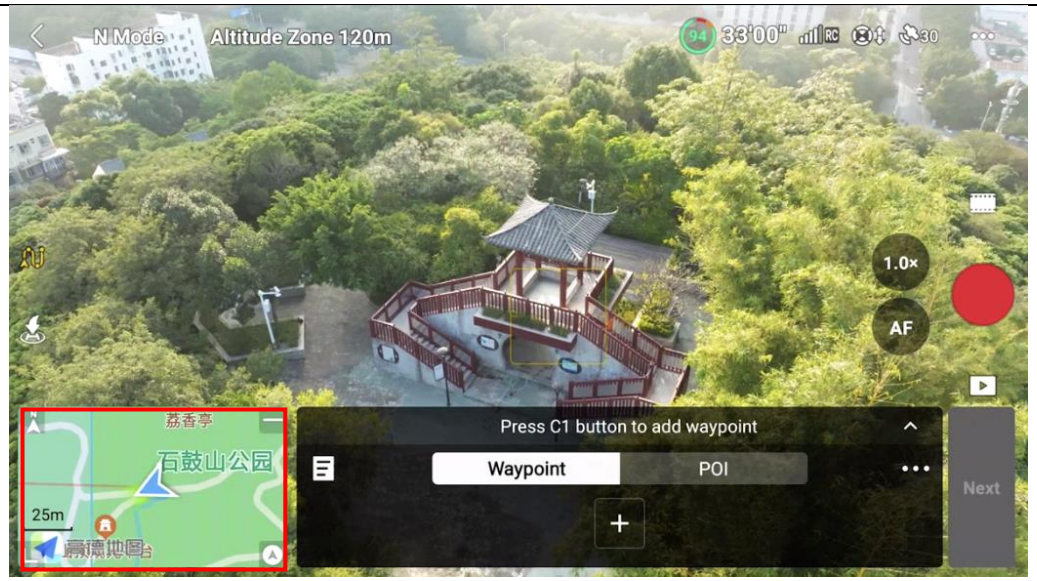
1. After the aircraft takes off, pin waypoints by using the remote controller or operating on the camera view:

• Remote control: enter the current Waypoint operation panel, press the remote control Fn button, or the C1 button if you are using DJI RC.

• The operation interface: In the Waypoint operation panel, tap the plus sign.

• After the Waypoint is marked, the aircraft will record the flight altitude, nose direction, gimbal pitch angle, and other data at this point.





2. Pin waypoints via the map: Enter the map view. On the Waypoint Operation Panel, tap different locations on the map to pin multiple waypoints.

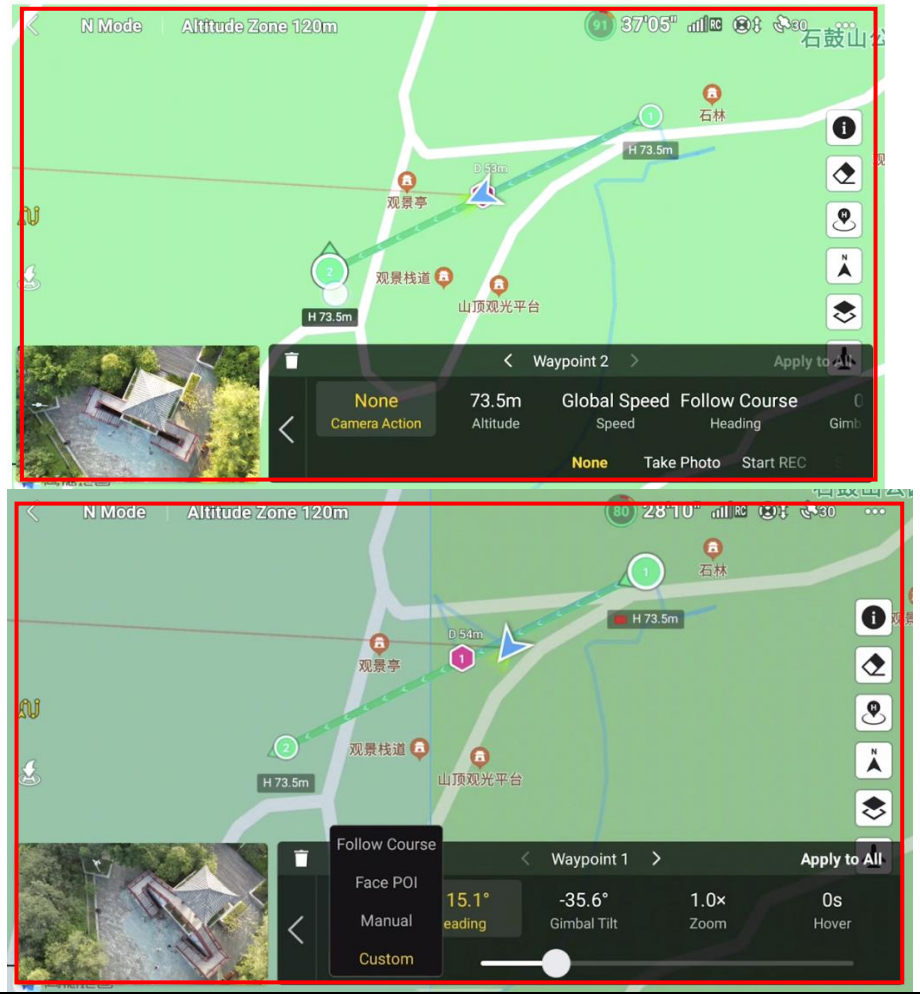
3. Change a waypoint location: Press and hold the specified waypoint icon on the flight map, then drag it to move the waypoint.

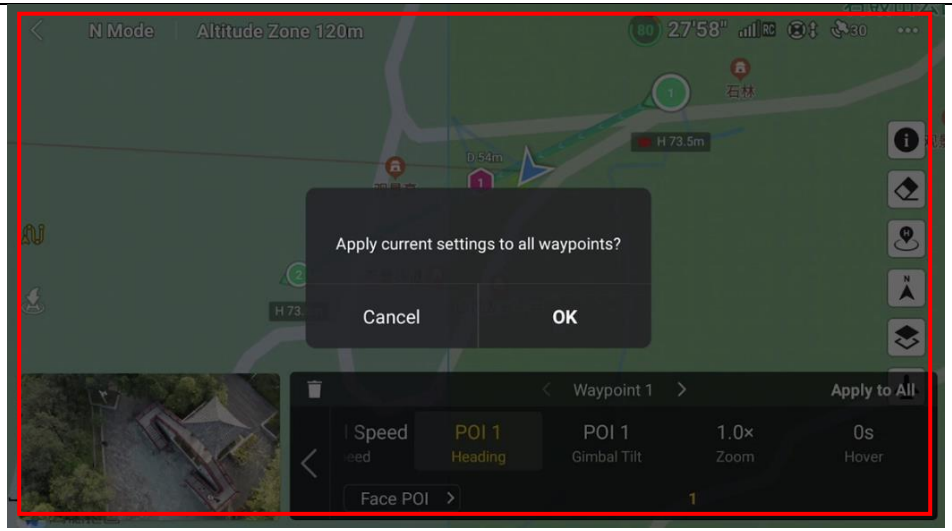
4. You can also plan waypoints on the map without connecting to the aircraft: Run the DJI Fly app, and tap the flight map to enter the flight map view. Pin target waypoints on the flight map and save the Waypoint settings. After connecting to the aircraft, you can apply the saved waypoints for flight.

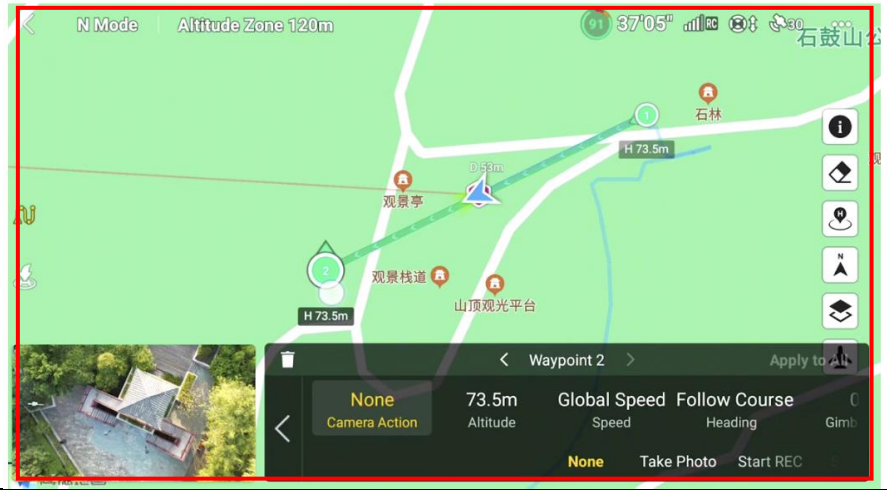
Waypoint Settings

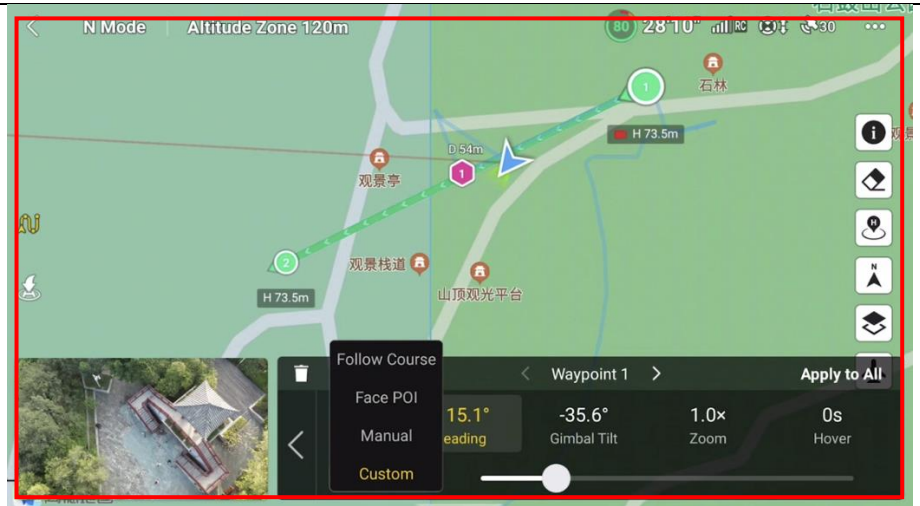
1. On the Waypoint Operation Panel, tap the specified waypoint number to enter the Waypoint settings. Tap the left or right angle brackets (< and >) in the title bar to switch between

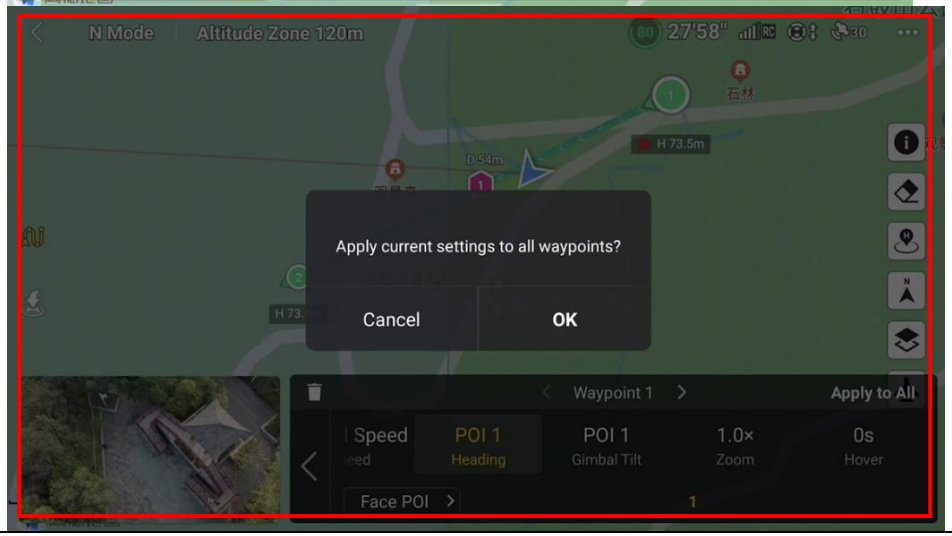
waypoints.



	 2. Waypoint settings include camera movement, altitude, speed, nose orientation, gimbal pitch, zoom mode, and hover time. • Camera Action: You can select Take Photo, Start Recording, or Stop Recording. • Altitude: You can customize the altitude of the aircraft relative to the takeoff point. • Speed: You can set the flight speed of the aircraft to Global Speed or Custom. When Global Speed is selected, the aircraft will fly at the same speed during the waypoint flight route. When Custom is selected, the aircraft will accelerate or decelerate at a steady speed when flying between waypoints, and the preset speed will be reached when the aircraft is at the waypoint. • Heading: You can select Follow Course, POI, Custom, or Manual. Custom: Drag the bar to change the orientation angle, and preview the orientation of the corner mark in the map. Manual: You can manually adjust the heading during flight. In this mode, the orientation of the aircraft nose is under your control. Source: https://support.dji.com/help/content?customId=en-us03400007343&spaceId=34&re=US&lang=en&documentType=artical&paperDocType=paper
receiving from the at least one user the	*The DJI Fly App receives the identification of a bookmarked area of the map (e.g., Point of interest) from the user.*

identification of a bookmarked area of the map;	*User can utilize the DJI Fly App to access the map in the application and can identify areas in the map to set a point of interest in the map.* **Waypoint Settings** 1. On the Waypoint Operation Panel, tap the specified waypoint number to enter the Waypoint settings. Tap the left or right angle brackets (< and >) in the title bar to switch between waypoints. 


N Mode
Altitude Zone 120m
石林
H 73.5m
观景亭
观景栈道
山顶观光平台
H 73.5m
Follow Course
Face POI
Manual
Custom
Waypoint 1
Apply to All
15.1°
-35.6°
Gimbal Tilt
1.0×
Zoom
0s
Hover


N Mode
Altitude Zone 120m
石林
H 73.5m
Apply current settings to all waypoints?
Cancel
OK
Waypoint 1
Apply to All
Speed
POI 1
Heading
POI 1
Gimbal Tilt
1.0×
Zoom
0s
Hover
Face POI

2. Waypoint settings include camera movement, altitude, speed, nose orientation, gimbal pitch, zoom mode, and hover time.

• Camera Action: You can select Take Photo, Start Recording, or Stop Recording.

• Altitude: You can customize the altitude of the aircraft relative to the takeoff point.

• Speed: You can set the flight speed of the aircraft to Global Speed or Custom.

When Global Speed is selected, the aircraft will fly at the same speed during the waypoint flight route. When Custom is selected, the aircraft will accelerate or decelerate at a steady speed when flying between waypoints, and the preset speed will be reached when the aircraft is at the waypoint.

• Heading: You can select Follow Course, POI, Custom, or Manual.

Custom: Drag the bar to change the orientation angle, and preview the orientation of the corner mark in the map.

Manual: You can manually adjust the heading during flight. In this mode, the orientation of the aircraft nose is under your control.

Set a POI

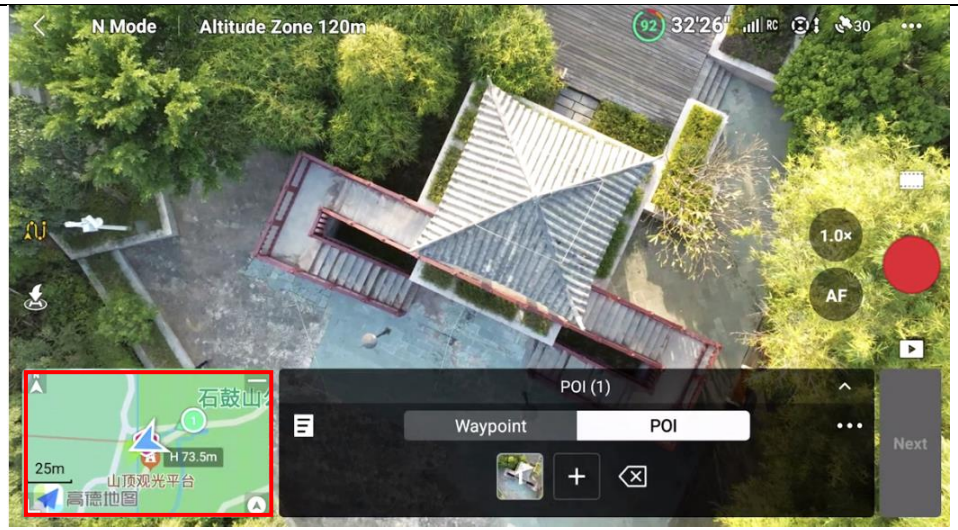
About Points of Interest (POI): If you associate a waypoint with a POI, the aircraft will face the POI while flying.

1. To capture toward a specific point, you can pin and set POIs. You can use the same method to pin a POI as used with a waypoint:

Tap POI to enter the POI Operation Panel. As with waypoints, you can pin POIs using the remote controller, or operating on the camera view or the flight map.

When adding POIs on the flight map, the default altitude of POIs is 50 meters.

2. To change a POI location, press and hold the specified POI icon on the flight map, then drag it to move the POI.

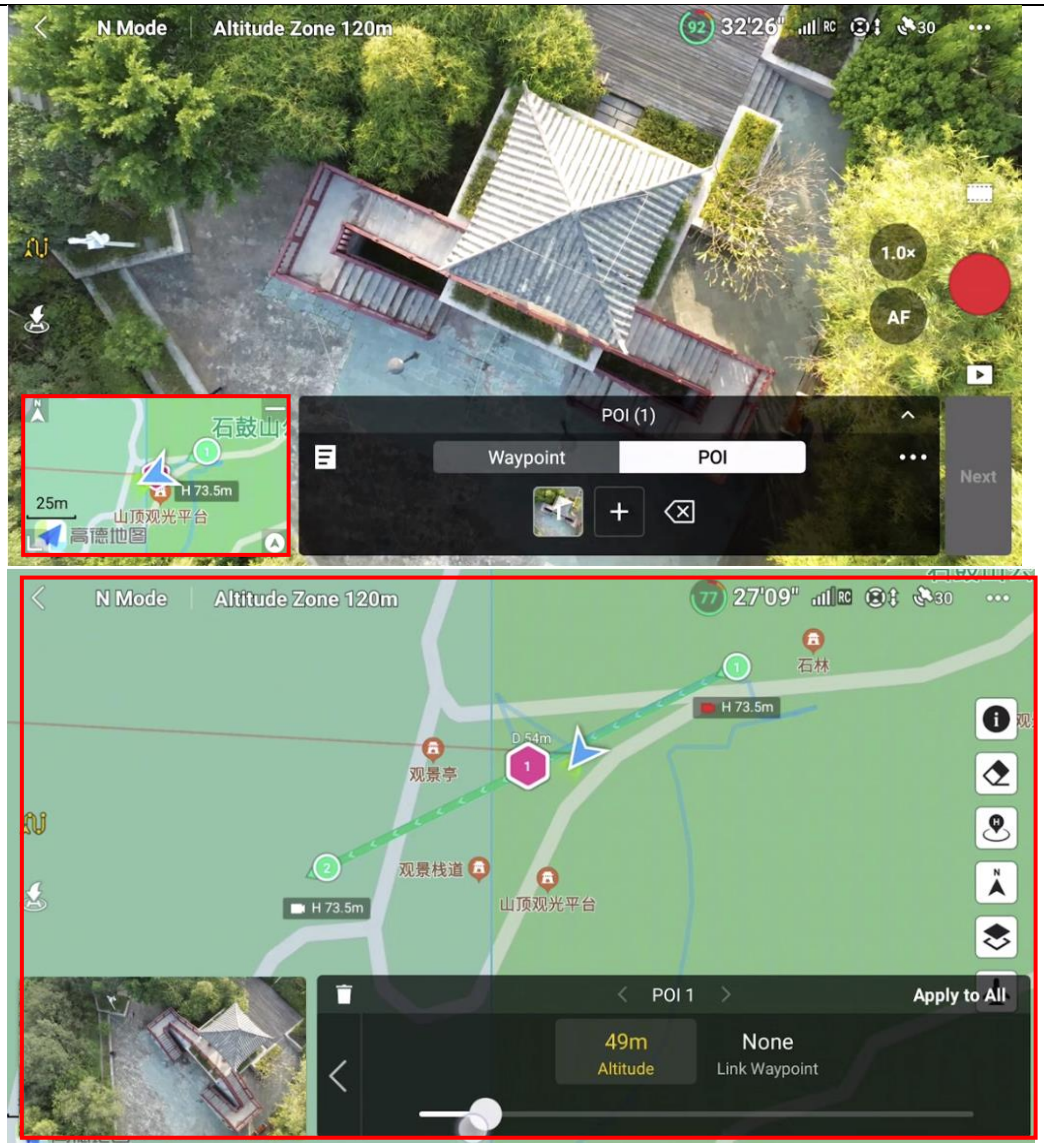


POI Settings

1. On the POI Operation Panel, tap the number of POI you want to set to carry out settings for an individual POI. Tap the left or right angle brackets (< and >) in the title bar to switch between POIs.

2. In POI Settings, you can set the altitude of and link waypoints to a POI.

In the Link Waypoint Settings, multiple waypoints can be linked to the same POI.


N Mode
Altitude Zone 120m
32'26"
1.0×
AF
POI (1)
Waypoint
POI
Next
石鼓山
H 73.5m
25m
山顶观光平台
高德地图
N Mode
Altitude Zone 120m
27'09"
石林
H 73.5m
观景亭
观景栈道
山顶观光平台
H 73.5m
POI 1
Apply to All
49m
Altitude
None
Link Waypoint

	Source: https://support.dji.com/help/content?customId=en-us03400007343&spaceId=34&re=US&lang=en&documentType=artical&paperDocType=paper
connecting the user to a decision loop associated with a selected mission; and creating model instances necessary to automatically provides alerts to the at least one mobile device regarding the bookmarked area.	*DJI Fly app connects the user to a decision loop (e.g., able to manage and control the operations of the unmanned aerial vehicle) associated with a selected mission (e.g., route or waypoint created).* *Creating model instances (e.g., save route and its tasks) necessary to automatically provides alerts (e.g., automatically capture images or videos of the location) to the at least one mobile device (e.g., Android or iOS) regarding the bookmarked area (e.g., point of interest).* *The user of the DJI Fly app can create waypoints and marks point of interest in the map. Based on the point of interest marking, the DJI Fly app connected with the unmanned aerial vehicle collects data related to the videos or images from the point of interest (e.g., bookmarked area).*  Source: https://www.dji.com/global/dji-fly

New Point of Interest (POI) flight mode

Point of Interest is a new intelligent flight mode that will allow you to quickly orbit a point of interest and collect comprehensive data with a 360-degree view of the target. This has been one of the most requested features by first responders and public safety agencies who want to easily add an orbit to their accident reconstruction data capture or orbit a scene. Vertical structures or areas can also be captured utilizing this tool.

Source: https://enterprise-insights.dji.com/blog/march-2023-enterprise-firmware-update

Set a POI

About Points of Interest (POI): If you associate a waypoint with a POI, the aircraft will face the POI while flying.

1. To capture toward a specific point, you can pin and set POIs. You can use the same method to pin a POI as used with a waypoint:

Tap POI to enter the POI Operation Panel. As with waypoints, you can pin POIs using the remote controller, or operating on the camera view or the flight map.

When adding POIs on the flight map, the default altitude of POIs is 50 meters.

2. To change a POI location, press and hold the specified POI icon on the flight map, then drag it to move the POI.

Source: https://support.dji.com/help/content?customId=en-us03400007343&spaceId=34&re=US&lang=en&documentType=artical&paperDocType=paper

Creating model instances to automatic provide alerts (i.e., capture images or videos related to POI)


1.3×
AF
40°
04:51
USING THE SCREEN ON EITHER THE CONTROLLER OR DEVICE DRAW A BOX AROUND THE AREA TO CIRCLE


1.3×
AF
04:52
THE DRONE WILL THEN ADJUST ITS POSITION AND THE TRACKING MENU SHOWS



Automatically capturing images or videos of a point of interest and providing it to the user

Source: https://www.youtube.com/watch?v=ZArVJpwWqAg

Saving a Route

1. Tap to open the route saved in history.

2. Tap the pencil icon to modify the route task name.

3. You can swipe the waypoint flight to the left and tap Delete to delete it.

4. Tap the icon in the upper right corner to change the sorting method of routes.

• Time icon: routes will be sorted in the order of save time.

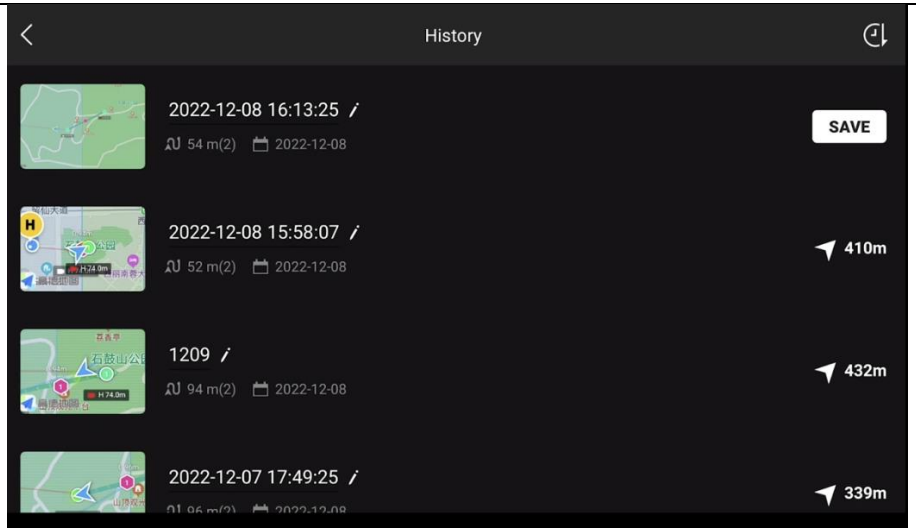
• Scale icon: The routes will be sorted by the distance from the current position to the starting Waypoint from shortest to farthest.

💡 **[Tips]** Currently, Waypoints flight missions cannot be exported and uploaded to the server, and can only be called from the DJI Fly app on the mobile device where the missions are saved.

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Creating and saving a model or profile to perform a mission and capture images or videos related to the point of interest



Source: https://support.dji.com/help/content?customId=en-us03400007343&spaceId=34&re=US&lang=en&documentType=artical&paperDocType=paper